Bowen Li

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EDUCATION

University of Southern California Los Angeles, CA
M.S. Industrial and Systems Engineering – Analytics ; GPA:3.61/4.00 Aug 2022 – May 2024

Dalian Polytechnic University Dalian, CHN
B.S. Computer Science and Technology; GPA:3.40/4.00 Aug 2018 – May 2022

SKILLS

Sports Analytics & Modeling: MixSort, YOLOX, CLIP-ReID, RTMPose, LightGBM, Archetype Analysis, GLMM, GAM, PyMC
Programming & Tools: Python (Pandas, NumPy, scikit-learn, PyTorch, OpenCV), R, Shell, MySQL/SQLite, Git, GCP, Streamlit

WORKING EXPERIENCES

Athleo.ai **Remote**
Developer of Basketball Video Auto-Clipping System Jan 2026 –Present

- **Pipeline Architecture:** Designed an end-to-end auto-clipping workflow for basketball footage using Python, PyTorch, and OpenCV, enabling user-specific clip generation based on selected players in under 30 seconds per game segment, reducing manual editing time by 80%.
- **Detection & Tracking:** Applied YOLOX with MixSort to process raw game footage, producing per-frame player IDs and bounding boxes with >85% tracking accuracy, ensuring reliable downstream analysis
- **Player Re-Identification:** Implemented CLIP-based embedding comparison with cosine similarity to link player appearances into tracklets, improving clip segmentation accuracy.

Cognizant – GOOGLE XR ML Data Collection & Annotation **Los Angeles, CA**
Data Analyst Jan 2025 – Present

- **Data Visualization:** Designed a Tableau dashboard analyzing 10,000+ survey responses across 20 projects, providing insights on user satisfaction, facilitator performance, and equipment usability. Utilized n-gram word clouds, drill-down analysis, and network diagrams to uncover recurring device issues, leading to hardware improvements and streamlined data collection processes.
- **Automation & Scripting**: Automated CSAT survey proctor name updates using Google Sheets, App Script, and SQL, ensuring real-time synchronization and reducing manual workload. Eliminated 5+ hours of manual interventions per week, enhancing data accuracy and operational efficiency across multiple projects.
- **Cross-functional Collaboration**: Partnered with hardware engineers, project leads, and research participants to execute high-volume VR headset data-collection sessions for face-tracking research, ensuring >98% session data integrity.

SPORTS ANALYTICS EXPERIENCE

[League-Wide NBA Player Archetype Portal - Brooklyn Nets (Interview Take-Home)] Jul 2026 – Aug 2026

- **Data Pipeline**: Consolidated league-wide NBA play-by-play, box-score, and Synergy play-type data into SQLite across all 30 teams and a full 1,230-game season — parsing raw play-by-play into 75,587 possession-level stints as the foundation for every downstream model.
- **Modeling**: Fit an 8-archetype Archetypoid Analysis model over all 433 qualifying NBA players — describing every player as an interpretable blend of eight *real-player* archetypes for league-wide role comparison.
- **Rookie Projection:** Built a Dirichlet regression translating a college stat line into a projected NBA archetype for zero-NBA-history rookies — validated on a held-out draft class at 52.8% top-match accuracy vs. a 41.7% baseline.
- **Scouting Dashboard**: Delivered a deployed multi-page Streamlit portal with an interactive 3D archetype map, per-player usage-vs-production diagnostics, and one-click PDF reports — putting league-wide player evaluation into a front-office-ready tool.

[NBA Draft 2026 – Shot Quality Scouting Portal] Apr 2026 – May 2026

- **Data Pipeline**: Built an end-to-end NCAA shot data pipeline via CBBD API, ingesting shot-by-shot records for 65 draft prospects into SQLite with automated box score, bio, and international stat collection — covering the full data foundation for prospect evaluation.
- **Modeling & CV Pipeline**: Trained a LightGBM xPTS model with Empirical Bayes shrinkage to compute PAE/100 per prospect; built a YOLOv11n jump-shot classifier (mAP50 = 0.828) and integrated RTMPose for frame-level joint angle extraction across 20+ clips per player.
- **Scouting Dashboard**: Delivered a 4-page Streamlit portal featuring a sortable draft board, per-player dossiers with Plotly radar charts and half-court zone diagrams, head-to-head comparison, and film-based mechanics analysis — enabling end-to-end prospect evaluation from raw shot data to video mechanics.

[Catcher Framing Modelling Project - Arizona Diamondbacks (Interview Take-Home)] Nov 2025 – Dec 2025

- **Modeling**: Isolated catcher contribution to ball/strike calls using a hierarchical mixed-effects model with a LightGBM probability baseline (5-fold CV, AUC = 0.98), engineered from spatial and count-context pitch-tracking features.
- **Pipeline**: Built a production scoring pipeline that loads the trained model, scores incoming pitch-level data, and auto-generates year-by-catcher framing summaries.